Task 6: Create a Strong Password and Evaluate Its Strength

1. Objective

To understand what makes a password strong and evaluate its strength using an online password strength checker.

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2. Tool Used

The tool used for this task is **passwordmeter.com**, which analyzes password strength based on length, complexity, and character variety.

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3. Password Tested

12345 – 10% – Very Weak | Tamil@123 – 60% – Medium | T@m1l#2026Secure – 100% – Strong

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4. Analysis

The results show that simple passwords like "12345" are very weak because they are short and contain only numbers, making them easy to crack using brute force attacks.
Passwords like "Tamil@123" are moderately strong but still predictable because they contain common words and patterns.
The strongest password "T@m1l#2026Secure" includes uppercase letters, lowercase letters, numbers, and symbols, along with good length. This makes it highly secure and difficult to break using brute force or dictionary attacks.

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5. Common Password Attacks

Brute Force Attack:
In this method, attackers try all possible combinations until the correct password is found.

Dictionary Attack:
This method uses a list of common passwords and words to guess the password quickly.

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6. Best Practices for Strong Passwords

- Use at least 12–16 characters
- Include uppercase and lowercase letters
- Add numbers and special symbols
- Avoid common words and personal information
- Use different passwords for different accounts
- Enable two-factor authentication (2FA)

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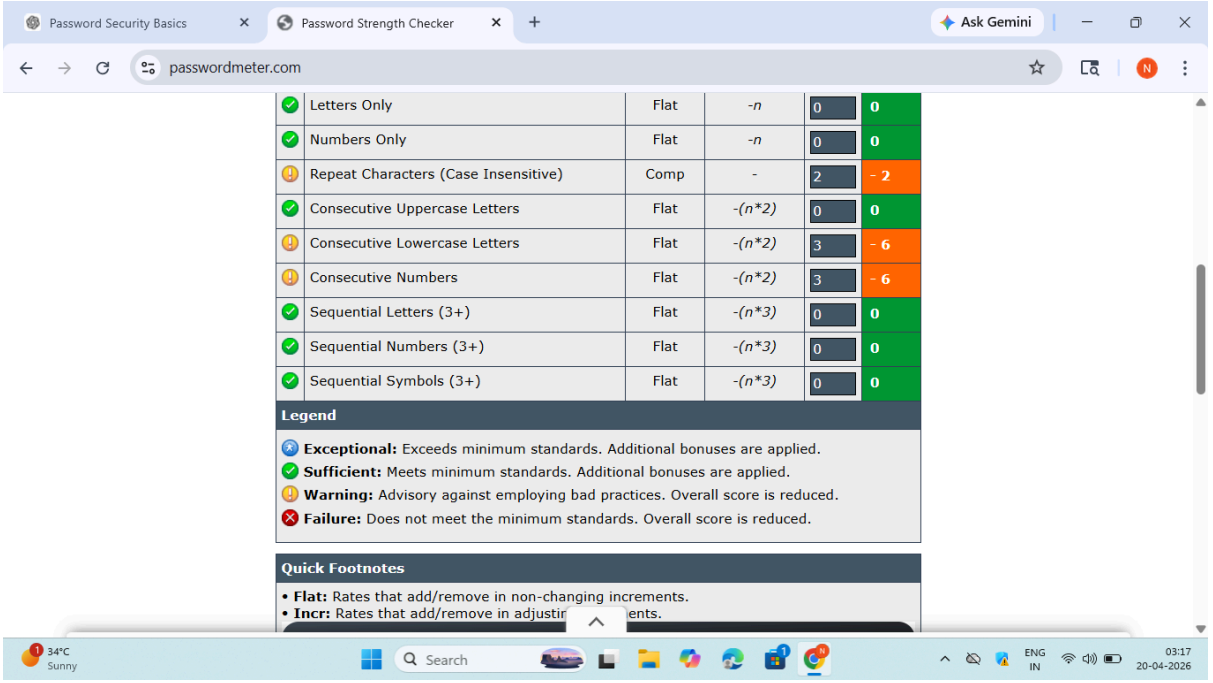
7. Screenshot

Strong Password:



Letters Only	Flat	-n	0	0
Numbers Only	Flat	-n	0	0
Repeat Characters (Case Insensitive)	Comp	-	2	- 2
Consecutive Uppercase Letters	Flat	-(n*2)	0	0
Consecutive Lowercase Letters	Flat	-(n*2)	3	- 6
Consecutive Numbers	Flat	-(n*2)	3	- 6
Sequential Letters (3+)	Flat	-(n*3)	0	0
Sequential Numbers (3+)	Flat	-(n*3)	0	0
Sequential Symbols (3+)	Flat	-(n*3)	0	0

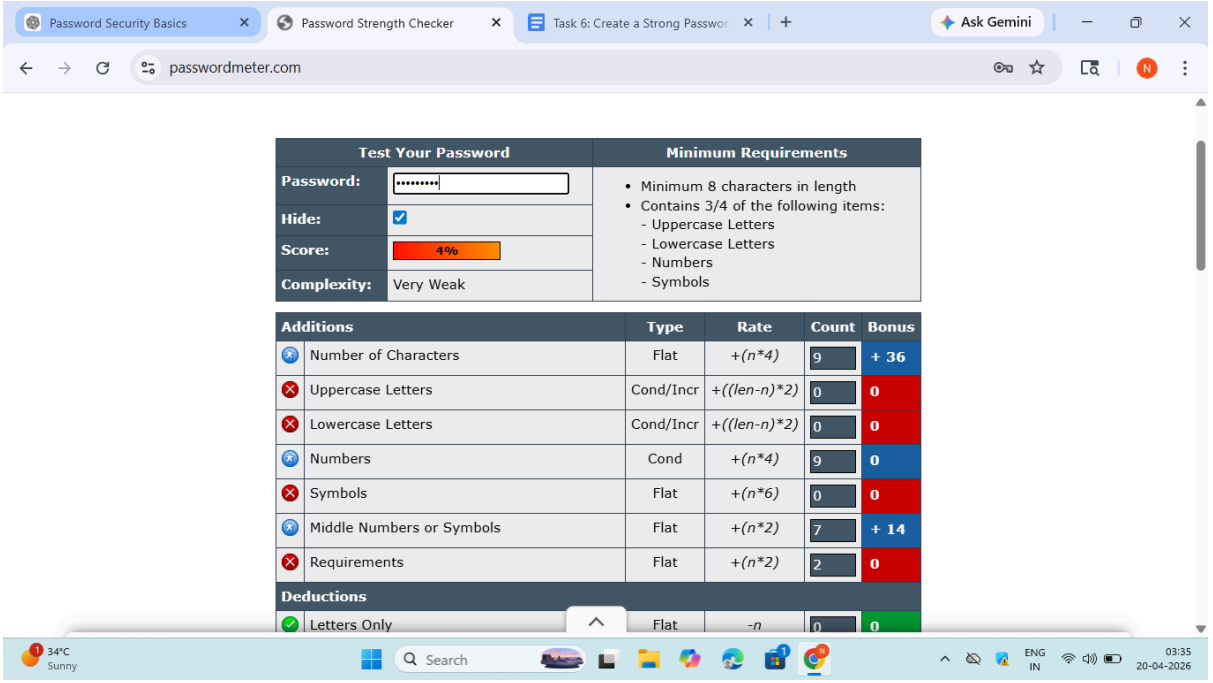
Legend

- **Exceptional:** Exceeds minimum standards. Additional bonuses are applied.
- **Sufficient:** Meets minimum standards. Additional bonuses are applied.
- **Warning:** Advisory against employing bad practices. Overall score is reduced.
- **Failure:** Does not meet the minimum standards. Overall score is reduced.

Quick Footnotes

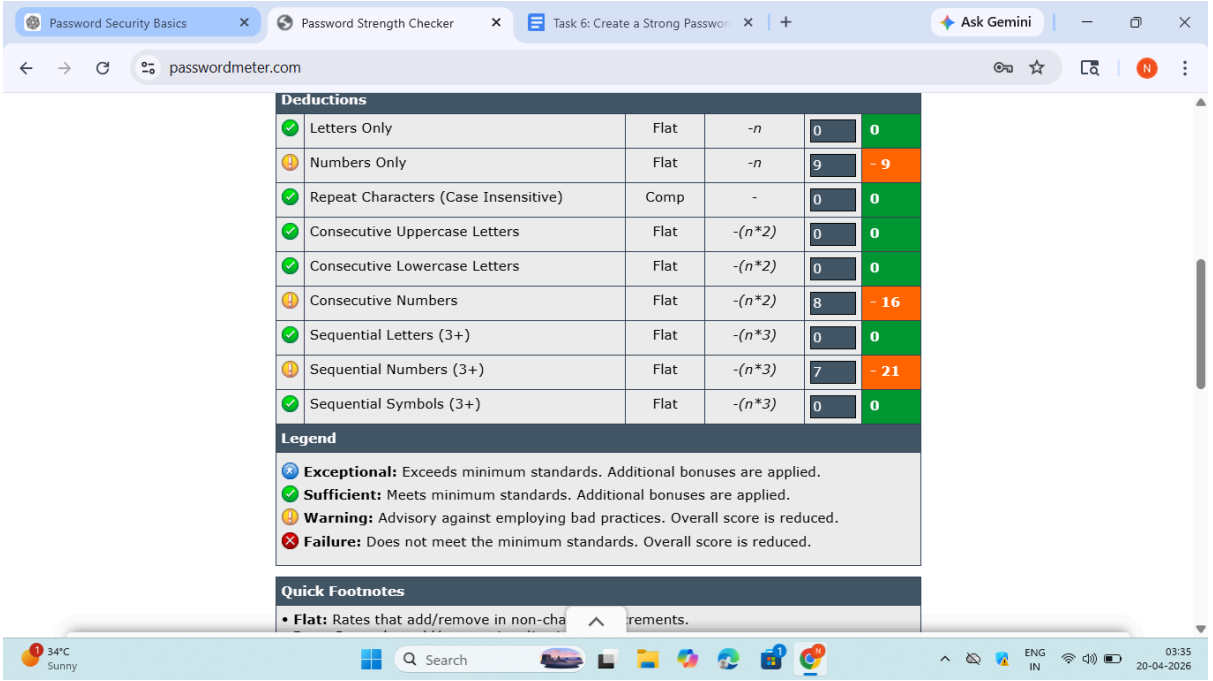
- **Flat:** Rates that add/remove in non-changing increments.
- **Incr:** Rates that add/remove in adjustin ents.

Weak Password:



Test Your Password		Minimum Requirements
Password:	••••••••	• Minimum 8 characters in length • Contains 3/4 of the following items: - Uppercase Letters - Lowercase Letters - Numbers - Symbols
Hide:	☑	
Score:	4%	
Complexity:	Very Weak	

Additions	Type	Rate	Count	Bonus
Number of Characters	Flat	+(n*4)	9	+ 36
Uppercase Letters	Cond/Incr	+((len-n)*2)	0	0
Lowercase Letters	Cond/Incr	+((len-n)*2)	0	0
Numbers	Cond	+(n*4)	9	0
Symbols	Flat	+(n*6)	0	0
Middle Numbers or Symbols	Flat	+(n*2)	7	+ 14
Requirements	Flat	+(n*2)	2	0
Deductions				
Letters Only	Flat	-n	0	0



Deductions

Letters Only	Flat	-n	0	0
Numbers Only	Flat	-n	9	- 9
Repeat Characters (Case Insensitive)	Comp	-	0	0
Consecutive Uppercase Letters	Flat	-(n*2)	0	0
Consecutive Lowercase Letters	Flat	-(n*2)	0	0
Consecutive Numbers	Flat	-(n*2)	8	- 16
Sequential Letters (3+)	Flat	-(n*3)	0	0
Sequential Numbers (3+)	Flat	-(n*3)	7	- 21
Sequential Symbols (3+)	Flat	-(n*3)	0	0

Legend

Exceptional: Exceeds minimum standards. Additional bonuses are applied.
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Warning: Advisory against employing bad practices. Overall score is reduced.
Failure: Does not meet the minimum standards. Overall score is reduced.

Quick Footnotes

• **Flat:** Rates that add/remove in non-cha rements.

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8. Conclusion

This task helped in understanding the importance of strong passwords. Weak passwords can be easily hacked, while strong passwords provide better security. Using complex and unique passwords is essential to protect personal data and accounts.